

Ordering one of our handcrafted frames has become so much simpler! Since our company only creates frames for pictures with integer-length sides, we're taking advantage of one of the most famous equations and now you only need to enter a single number to describe your frame size! Simply square the width and the length, and add them together!

$$W^2 + L^2 = D^2$$

Send us the sum (D^2). We will do the math to find W and L for the largest area and start building!

$$\text{Area} = (W)(L)$$

Disclaimer: We want you to get the most for your order, so we build our frames with the largest area possible. If you enter a D^2 value that has multiple side combinations (W, L) to achieve it, we will choose the Width and Length to give you the largest area! If you want skinny frames, we'll take your call another time.

We need you to implement the math portion of our interface!

Input

The input is a single positive integer N (at most 20000), the square of the diagonal (D^2) of the rectangle.

50

Output

Find the Width (W) and Length (L) of the rectangle, with $W \leq L$. There may be more than one solution, so choose sides with the largest area. Print W and L on one line separated by a space. If you can't find two integer sides, print "That diagonal does not lead to integer sides!"

5 5

Discussion

In the example above, "1 7" also has a D^2 of 50, but "1 7" has an area of 7, and "5 5" has an area of 25, so "5 5" is the best answer.

Additional Examples

Input 2	Output 2	Input 3	Output 3
18049	25 132	18051	That diagonal does not lead to integer sides!

